GRADUATION PROJECT PROPOSAL · FULL WORKFLOW

Basmati · بصمتي

Daily carbon footprint for people and a government facility: a phone app and an IoT energy monitor, piloted on the Computer Science department floor

Team	[9 team members]
Supervisor	[Supervisor name]
Pilot site	Computer Science department floor, [Faculty / University] (a government facility)
Duration	20 weeks, starting early November 2026
Estimated budget	8,140 EGP
Contents	Idea, architecture, workflow, data flow, equations, AI models, datasets, performance metrics, related work 2020–2026 with links, plan and budget
Date	October 2026

1. Summary

Basmati (بصمتي, "my footprint") is an Android app and a small IoT system that calculate a daily carbon footprint, in kilograms of CO_2-equivalent, with almost no typing. The project is built for a government facility. The pilot is one floor: the Computer Science department.

It produces two numbers every day:

- **A personal footprint** for each member of the department who uses the app: their commute, their meals, and their share of the floor's electricity for the hours they spent there, compared with a daily budget of 6.8 kg.
- **A floor footprint** for the facility: the floor's measured electricity, its CO_2e, the energy used after working hours, and a forecast, shown on a display at the floor entrance and on a dashboard for the department head.

The project uses three AI models on real Egyptian data: transport mode detection, meal recognition from a photo, and a forecast of the floor's electricity. A fourth model detects after-hours waste. Two ESP32 monitors with current clamps measure the floor's electrical board. Every model is compared with a simple baseline, and every target is measured on people and days that were not used for training.

2. Problem and motivation

- **No idea of size or source.** Neither people nor facility managers know how much CO_2e they produce, or which habit or room produces most of it.
- **One bill, too late.** A government building receives one monthly bill for the whole building. It does not show the floor, the hour or the person, and it arrives after the energy is used. Egyptian university buildings usually lack even monthly data per building [23].
- **Where the energy goes.** In Cairo's public administrative buildings about 40% of electricity goes to cooling and 60% to lighting and equipment [24]. In an Egyptian faculty building, PCs, laboratories, lighting and air conditioning dominate [23]. ACs and lab PCs left on after hours are invisible.
- **Not built for Egypt.** Personal carbon calculators are yearly questionnaires with foreign factors (no microbus, no koshary, no Egyptian grid), and they give inconsistent results [29].

Goal: measure the CS floor and each member's day automatically, with Egyptian factors and real data, and show where the waste is.

3. Background: carbon footprint and the daily budget

A carbon footprint is the total greenhouse gas caused by a person, product or building over a period, measured in kg of CO_2-equivalent (CO_2e). Every activity converts to CO_2e through an emission factor: kg CO_2e per kWh of electricity, per km travelled or per kg of food.

Figure	Value	Source
Egypt, per person per year (2023)	3.4 t CO_2e	EDGAR 2024 report [27]
World average, per person per year (2023)	6.6 t CO_2e	EDGAR 2024 report [27]
1.5 °C lifestyle target per person by 2030	2.5 t CO_2e	Hot or Cool Institute [26]
Daily budget per person	2,500 ÷ 365 ≈ 6.8 kg	Our calculation

Personal colors. A day is green up to 4.8 kg (70% of the budget), yellow up to 6.8 kg, and red above 6.8 kg.

Floor colors. The floor is compared with its own baseline (the same weekday in the first three weeks, before any feedback): green when it uses at most 90% of the baseline, yellow up to 100%, red above.

Country figures are territorial emissions; the target is a lifestyle (consumption) figure, so the comparison is indicative.

4. Proposed system

4.1 Components

Component	What it does
Android app (Kotlin, Jetpack Compose)	Detects trips and transport mode; recognizes meals from a photo; counts hours on the floor with a geofence check-in; shows the daily total, color and tips; has a volunteer labeling mode
Floor monitors (2 × ESP32)	Monitor A: a CT clamp on each of the three main phases of the floor's electrical board. Monitor B: clamps on three circuits (lab ACs, lab PCs, lighting). Send readings every 10 s over MQTT
Server (FastAPI)	Receives trips, meals and readings; runs the daily footprint engine, the floor forecast and the waste detector; stores data in PostgreSQL with TimescaleDB
Entrance display	An LED ring at the floor entrance shows the floor's day in green, yellow or red
Facility dashboard	Daily kWh, CO_2e, after-hours waste, forecast and a weekly report for the department head; waste alerts to the floor technician

4.2 Pilot site and approvals

The pilot is the Computer Science department floor. Before installation the team needs an official approval from the faculty administration, access to the floor's electrical board, the department timetable and the building's past electricity bills. In week 2 the team surveys the floor: rooms, laboratories, number of PCs and ACs, and the circuits on the board, following the datasheet idea of Safwat et al. [23].

Safety. CT clamps close around the cables without cutting them, and the monitors run on 5 V adapters, so no mains voltage enters their circuits. The faculty electrician installs them with the board switched off. The first test is done in the lab on a known load, such as a 2,000 W kettle.

4.3 Scope

In scope	Out of scope
Commute, meals and the floor's electricity for department members; optional home part from the monthly bill	Purchases, flights (manual entry only), other floors and buildings
Measuring, forecasting and alerting on the floor's electricity	Switching devices on or off (we measure and alert; we do not control)
Android app, server, two monitors, dashboard	iOS version

5. System architecture

The phones and the floor monitors send small records to one server, which adds up the day. The models run on the phone, so photos and locations never leave it: the server receives only final numbers such as distance, mode, dish, portion and hours on the floor.

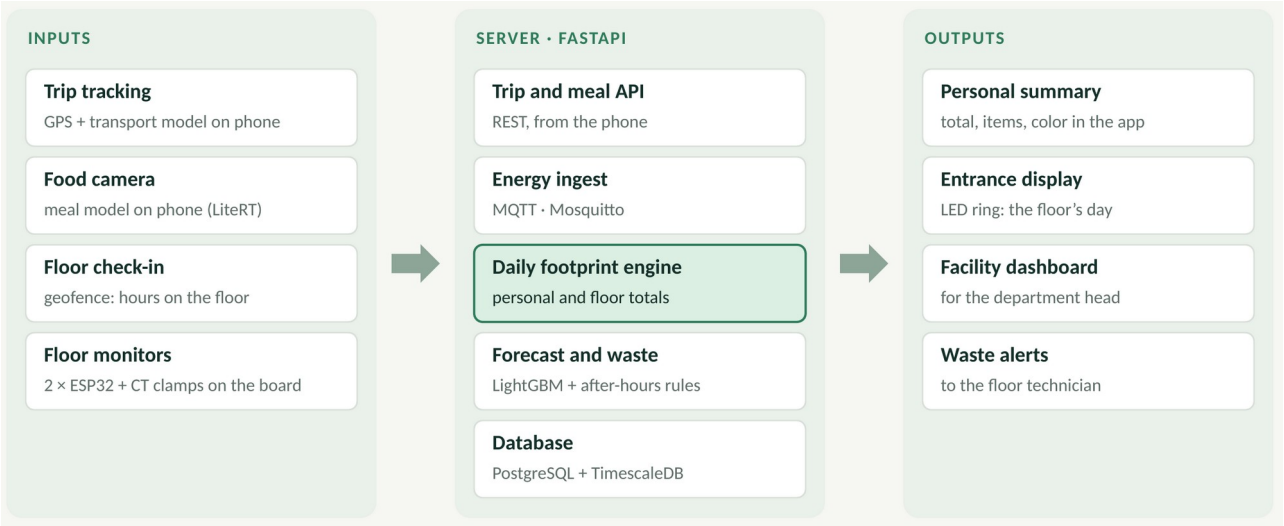


Figure 1. System architecture: inputs, server and outputs.

Layer	Technology
Mobile	Kotlin, Jetpack Compose, CameraX, LiteRT (TensorFlow Lite), Room, Activity Recognition API, geofencing
Hardware	ESP32, SCT-013 current clamps (100 A on main phases, 30 A on circuits), ADS1115 16-bit ADC, WS2812B LED ring
Messaging	MQTT with the Mosquitto broker; REST/JSON from the phone
Server	Python, FastAPI, PostgreSQL + TimescaleDB, LightGBM, scikit-learn
Dashboard	Web dashboard (Grafana or a simple web page) on a faculty server or a free cloud tier

6. Workflow

The app senses, the user confirms only when needed, and the server adds up. Every answer from a user becomes new training data.



Figure 2. The six steps of the workflow.

1. **Sense:** the phone records GPS and motion sensors, takes the meal photo and checks in at the department; the monitors sample current.
2. **Classify:** on-phone models give the transport mode and the top 3 dishes; the server turns current into kWh.
3. **Confirm:** the user answers with one tap only when the model is unsure (confidence below 70%) and picks the portion size.
4. **Compute:** the daily footprint engine applies the emission factors to each record (Section 8).
5. **Notify:** at 11 pm the app shows the personal total and color; the entrance display and dashboard show the floor's day; after-hours waste raises an alert.
6. **Improve:** confirmed modes, corrected dishes and confirmed waste alerts are added to the training data for the next model version.

When	What happens
A trip ends	Mode classified on the phone; one-tap question if confidence is below 70%
A meal photo is taken	Top 3 dishes shown; the user picks the dish and the size
Arriving at the department	Geofence check-in counts the hours on the floor (no location is kept)
Every 10 seconds	Floor monitors send current per phase and per circuit
After working hours	Load above the base level outside the timetable raises a waste alert
11 pm	Personal total and color in the app; the floor's total and color on the entrance display
Every week	A personal tip aimed at the biggest item; a floor report to the department head

7. Data flow

Three pipelines feed the daily footprint engine. Each sends small records, not raw media.

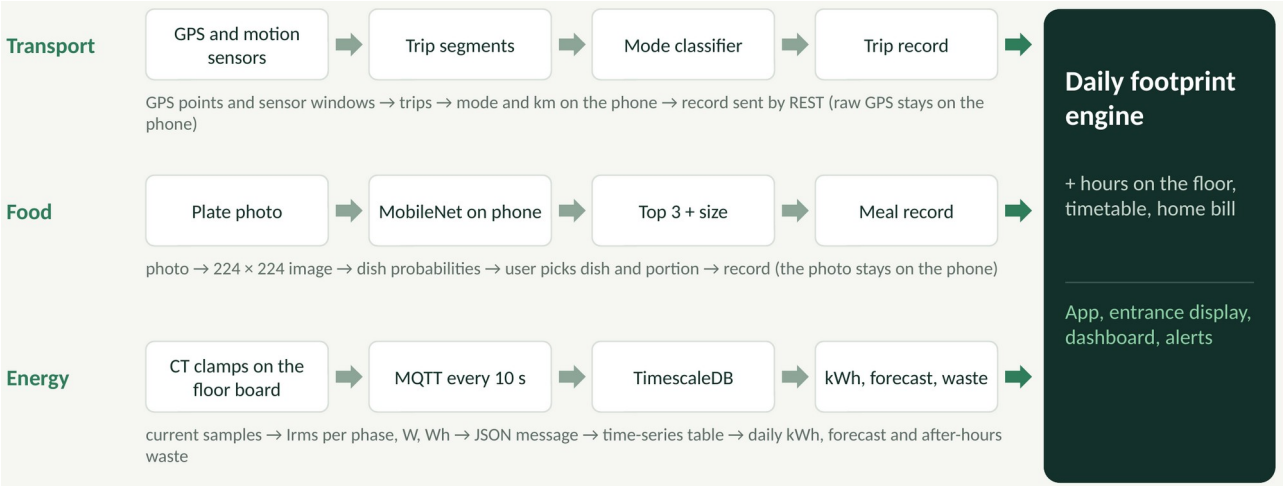


Figure 3. Data flow: transport, food and energy pipelines.

- **Transport:** GPS points and sensor windows → trip segments → mode classifier and distance on the phone → one trip record sent by REST. Raw GPS stays on the phone and is deleted after the trip is summarized.
- **Food:** plate photo → 224 × 224 image → dish probabilities from MobileNet on the phone → the user picks the dish and portion → one meal record. The photo stays on the phone.
- **Energy:** current samples → Irms per phase, power (W) and energy (Wh) on the ESP32 → JSON message over MQTT every 10 s → time-series table → daily kWh, forecast and after-hours waste on the server.
- **Other inputs:** hours on the floor (one number per user per day), the department timetable (once per term) and, optionally, the home electricity bill and gas cylinders (once per month).

7.1 Records

Record	Key fields	Rate	Stored in
Meter reading	device, circuit, time, Irms, W, Wh	every 10 s	TimescaleDB
Trip record	user, start, end, km, mode, confidence, passengers	per trip	PostgreSQL
Meal log	user, dish, portion, time	per meal	PostgreSQL
Floor presence	user, date, hours on the floor	daily	PostgreSQL
Timetable	room, day, hour, expected people	per term	PostgreSQL
Waste event	circuit, start, end, kWh, confirmed	when found	PostgreSQL
Daily footprint	user or floor, date, items, total, color	daily	PostgreSQL

Privacy by design. Photos and locations never leave the phone, presence is stored as hours only, volunteers are IDs instead of names, raw GPS is deleted once the trip is summarized, and every volunteer signs a consent form and can withdraw. Emission factors live in one settings file on the server, so a factor can be updated without changing code.

8. Equations

8.1 The core rule

$$\mathrm{CO_2e} = \sum_i A_i \times \mathrm{EF}_i$$

Every footprint is an activity amount A_i (kWh, km, kg) multiplied by its emission factor EF_i in kg CO_2e per unit.

Activity	Unit	kg CO_2e / unit	Source
Egypt grid electricity (2024)	kWh	0.45	Low-Carbon Power (447 g/kWh)
Petrol car, divided by passengers	km	0.17	UK DESNZ (approximate)
Microbus or bus	passenger-km	0.10	Our estimate
Metro	passenger-km	0.06	Our estimate
Walking or cycling	km	0	—
LPG cylinder (12.5 kg)	cylinder	37.5	LPG combustion (approximate)
Beef (dairy herd to beef herd)	kg	33 to 99.5	Poore and Nemecek (2018)
Chicken	kg	9.9	Poore and Nemecek (2018)
Rice	kg	4.5	Poore and Nemecek (2018)
Lentils and beans	kg	1.8	Poore and Nemecek (2018)

8.2 A person's day

$$D = \sum_{\text{trips}} d \cdot f_{\text{mode}} + \sum_{\text{meals}} f_{\text{dish}} \cdot s$$
$$+0.45 \cdot E_{\text{floor}} \cdot \frac{h_i}{H} + \frac{0.45 E_{\text{home}} + G}{N_{\text{home}}}$$

Symbol	Meaning
d	trip distance in km
f_{mode}	mode factor (car factor divided by passengers)
f_{dish}	meal footprint from its ingredients (Section 8.3)
s	portion size: small 0.7, medium 1, large 1.4
E_{floor}	floor kWh per day, measured by the monitors
h_i and H	your hours on the floor, and all person-hours on the floor that day (from the timetable)
E_{home}	home kWh per day from the monthly bill (optional)
G and N_{home}	home gas (cylinders per month × 37.5 ÷ 30) and household members

Worked example. A student's day:

Item	Calculation	kg CO_2e
Transport	20 km by microbus × 0.10	2.0
Food	beans, two eggs, rice and 150 g of chicken	2.6
Floor share	0.45 × 330 kWh × 6 h ÷ 1,480 person-hours	0.6
Home	(0.45 × 250 kWh ÷ 30 + 37.5 ÷ 30) ÷ 4 members	1.2
Total	**6.4 kg = 94% of 6.8 kg: above the 4.8 kg green line, so the day is yellow**	**6.4**

Swapping the chicken for 150 g of beef raises the day to about 9.9 kg, which is red.

8.3 Meals

$$f_{\text{dish}} = \sum_{\text{j}} w_j \cdot \text{EF}_j \qquad F_{\text{meal}} = f_{\text{dish}} \cdot s$$

w_j is the raw weight of ingredient j in kg, EF_j its factor from Poore and Nemecek (2018), and s the portion size chosen by the user.

Dish (medium plate)	Ingredients, raw weight	kg CO_2e
Koshary	rice 100 g, lentils 50, pasta 50, tomato 50, onion 30	0.73
Chicken with rice	chicken 150 g, rice 100	1.93
Beef with rice	beef 150 g, rice 100	5.4 to 15.4

The team builds this table for 20 dishes, including the faculty cafeteria menu, using three recipe sources per dish plus a plate weighed on a kitchen scale. Beef uses the dairy-herd factor (33) as a conservative value.

8.4 The floor

$$C_{\text{floor}} = 0.45 \cdot E_{\text{floor}} \qquad I = \frac{C_{\text{floor}}}{H} \qquad r = \frac{E_{\text{floor}}}{E_{\text{base}}}$$

$$W = \sum_{t \notin T_{\text{work}}} \max\left(0, P_t - P_{\text{base}}\right) \cdot \Delta t$$

Symbol	Meaning
C_{floor}	the floor's footprint in kg CO_2e per day
I	intensity: kg CO_2e per person-hour on the floor
r	ratio to the baseline; green r ≤ 0.9, yellow r ≤ 1.0, red r > 1.0
E_{base}	average kWh for the same weekday in the first 3 weeks (before feedback)
W	after-hours waste in kWh
T_{work} and P_{base}	timetable hours, and the night base load (servers, fridges)

Example. A working day of 330 kWh gives 0.45 × 330 ≈ 149 kg CO_2e. With 1,480 person-hours the intensity is about 0.10 kg per person-hour. If lab ACs run for three hours after the last lecture and add 40 kWh above the base load, W = 40 kWh, about 18 kg CO_2e of waste.

8.5 The energy monitor

$$I_{\mathrm{rms}} = \left(\frac{1}{N} \sum_{n=1}^{N} i_n^2 \right)^{1/2}$$

$$P = k \cdot V \cdot \sum_{p=1}^{3} I_{\mathrm{rms},\,p} \qquad E = \sum_{t} P \cdot \Delta t \qquad k = \frac{E_{\mathrm{ref}}}{E_{\mathrm{dev}}}$$

1. Monitor A has a clamp on each of the three main phases; monitor B has clamps on lab ACs, lab PCs and lighting.
2. The ADS1115 samples the clamps (up to 860 samples per second in total); the ESP32 computes Irms for each channel.
3. Power uses V = 220 V and a calibration factor k, updated weekly from a reference reading: the floor sub-meter if there is one, otherwise a handheld power meter from the electrical lab.
4. Target: weekly error below 10% against the reference.

9. AI models

Model	Input	Method	Runs on	Baseline
Transport mode (core)	Speed, stops, acceleration, sensor vibration, GPS loss	Activity Recognition API, then Random Forest or LightGBM; exported to Kotlin with m2cgen	Phone	Activity Recognition API alone
Meal recognition (core)	Plate photo	Transfer learning on MobileNetV3 or EfficientNet-Lite0; LiteRT int8	Phone	Most common dish
Floor forecast (core)	Hourly kWh, hour, weekday, temperature, timetable	LightGBM; LSTM if there is enough data	Server	Same hour last week
After-hours waste (core)	Load per circuit, timetable, forecast	Rules first, then Isolation Forest on forecast errors	Server	Fixed threshold
Appliance breakdown (optional)	Floor main power	NILM, e.g. sequence-to-point; checked against monitor B	Server	—

Rule: every model must beat its baseline, measured first, on people and days not used in training. Metro trips are found by a rule: GPS is lost underground and the trip starts and ends inside a geofence around a station.

First 15 dishes: ful, ta'meya, eggs, cheese, koshary, lentils, chicken with rice, beef with rice, kofta, mahshi, molokhia, béchamel pasta, chicken shawarma, fish and salad, plus the cafeteria menu.

10. Datasets

10.1 Public datasets

Dataset	Year	Contents	Our use
ImageNet	2012	About 1.28M training images, 1,000 classes	Pretrained MobileNetV3 weights
Food-101	2014	101,000 images of 101 dishes	Food pretraining and benchmark
SHL	2018	Phone sensor data, 8 transport and locomotion modes	Pretrain the transport classifier
GeoLife	2012	GPS trajectories from 182 users, part with mode labels	Benchmark trip features
Building Data Genome 2	2020	Hourly meter data of 1,636 non-residential buildings over 2 years	Pretrain and benchmark the floor forecast
UK-DALE	2015	Whole-house and appliance demand, 5 UK homes	NILM benchmark (optional)
Poore and Nemecek	2018	Global food life-cycle emission factors	Dish footprints
OpenStreetMap	—	Locations of Cairo metro stations	Metro geofences
Open-Meteo	—	Hourly temperature at the faculty	Forecast input

Public datasets do not contain microbuses, Egyptian dishes or Egyptian public buildings, so they are used for pretraining and benchmarking only. The final evaluation is always on the data we collect.

10.2 Data collected by the team

Data	Amount	How	Weeks
Floor electricity	6 channels every 10 s; about 2,350 hourly readings each	Monitors on the floor board; weekly reference reading	6–20
Timetable and presence	Room schedule; hours on the floor per volunteer	Department administration; app check-in	2–20
Building bills	12–24 months of the building's electricity bills	Official request to the faculty administration	2–4
Food photos	200+ per dish: 3,000 photos from 30 people	Volunteers photograph meals and pick the dish name	4–10
Dish recipes	20 dishes with ingredients in grams	3 recipe sources each, plus a weighed plate	2–5
Labeled trips	20 volunteers × 2 weeks ≈ 560 trips; 80+ per mode	Volunteer mode records GPS and sensors; user picks the mode	5–9
Waste events	About 30 labeled after-hours events	Staged with the technician (e.g. one AC left on for an hour), plus confirmed alerts	8–18

- **Split:** 70 / 15 / 15 by person (trips, photos) and by week (electricity), never by photo, trip or hour, so test results reflect new people and new days.
- **Ethics:** consent forms, IDs instead of names, raw GPS deleted after feature extraction, right to withdraw, and official approval for the floor monitors.

11. Performance metrics

11.1 Definitions

Classification (transport mode per trip; waste events found or missed). TP, FP and FN are true positives, false positives and false negatives. For transport they are computed per mode and averaged (macro F1); for waste a detection is correct if it overlaps a labeled event.

$$\text{Precision} = \frac{\text{TP}}{\text{TP+FP}} \qquad \text{Recall} = \frac{\text{TP}}{\text{TP+FN}} \qquad F_1 = \frac{2PR}{P+R}$$

Regression (floor forecast, monitor error, trip distance). MAPE is used for the daily floor forecast; MAE and RMSE for distance and monitor error.

$$\text{MAE} = \frac{1}{n}\sum \mid y_i - \hat{y}_i \mid \qquad \text{RMSE} = \left(\frac{1}{n}\sum (y_i - \hat{y}_i)^2\right)^{1/2} \qquad \text{MAPE} = \frac{100\%}{n}\sum \frac{|y_i - \hat{y}_i|}{y_i}$$

Top-k accuracy (meal recognition): the share of photos whose true dish is among the model's k best guesses. We report Top-1 and Top-3, because the app shows three guesses.

$$\text{Top-}k = \frac{1}{n}\sum_{i=1}^{n} 1[y_i \in \text{top}_k(x_i)]$$

11.2 Targets

Part	Metric	Target
Transport mode	Accuracy per trip (and macro F1)	At least 85%
Trip distance	Error vs. true distance	Below 10%
Meal recognition	Top-1 / Top-3 accuracy	At least 80% / 95%
Meal speed	Time on a mid-range phone	Under 1 second
Floor monitor	Weekly error vs. reference reading	Below 10%
Floor forecast	Daily MAPE	Below 15%, and beats last week
Waste detection	Recall / precision on labeled events	At least 90% / 80%
Battery	App use per day	Below 5%
Real data	Volunteers and site	20 volunteers, 1 floor, 14 weeks

11.3 Evaluation protocol

- Measure every baseline first, then each model, on the same held-out people and weeks.
- First evaluation at the first full version (week 10); final evaluation in week 18.
- Report real numbers even when they fall short of a target, with the confusion matrix and the error analysis.
- Battery: run the app for a full day on three phone brands and read Android battery statistics.

11.4 Curves we will report

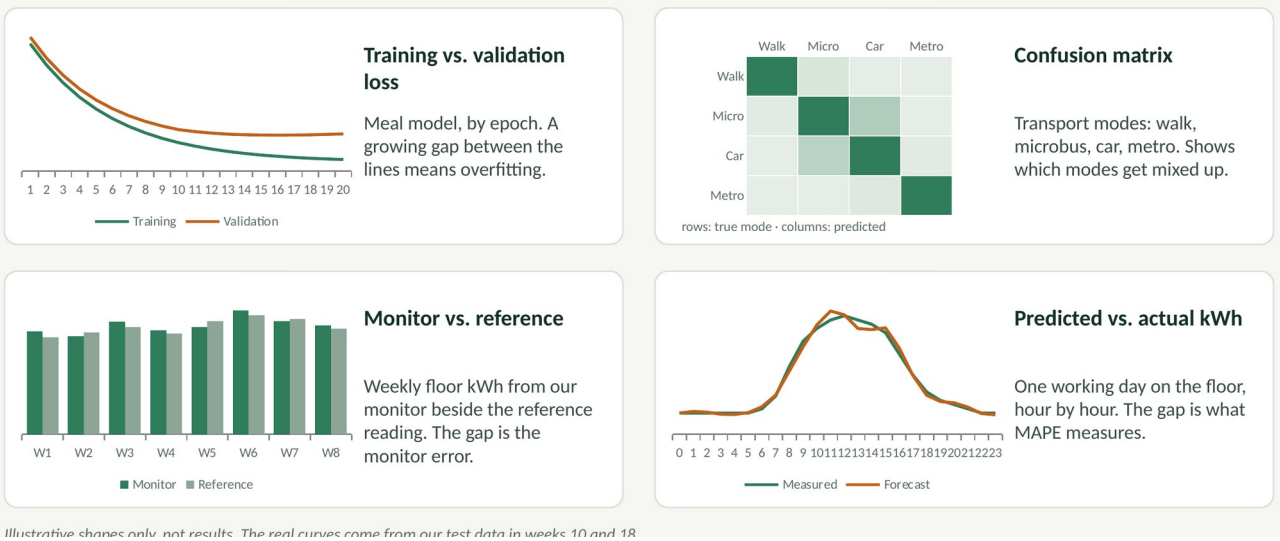


Figure 4. The four plots for the results chapter. Illustrative shapes only, not results.

- Training vs. validation loss of the meal model (overfitting check).
- Confusion matrix of transport modes (walk, microbus, car, metro).
- Weekly floor kWh: monitor vs. reference reading (monitor error).
- Predicted vs. actual hourly floor kWh on a working day (forecast error, MAPE).

12. Related work 2020–2026

We reviewed 40 papers and reports from 2020 to 2026. Most of them (references 1–25) are about how universities, colleges, schools and public buildings measure their carbon footprint; the rest support the targets and the technology of the app. The numbers match the reference list; Appendix A gives a link for each one.

#	Paper	Year	Group
1	Valls-Val and Bovea	2021	Measuring educational institutions: reviews, methods and tools
2	Li et al.	2021	Measuring educational institutions: reviews, methods and tools
3	Helmers, Chang and Dauwels	2021	Measuring educational institutions: reviews, methods and tools
4	Valls-Val and Bovea (CO2UNV)	2022	Measuring educational institutions: reviews, methods and tools
5	Guerrero-Lucendo et al.	2022	Measuring educational institutions: reviews, methods and tools
6	Herth and Blok	2023	Measuring educational institutions: reviews, methods and tools
7	Kiehle et al.	2023	Measuring educational institutions: reviews, methods and tools
8	Deda, Gervásio and Quina	2025	Measuring educational institutions: reviews, methods and tools
9	Abdelbasier et al.	2022	Measuring educational institutions: reviews, methods and tools
10	Yañez, Sinha and Vásquez	2020	Measured universities: case studies
11	Clabeaux et al.	2020	Measured universities: case studies
12	Filimonau et al.	2021	Measured universities: case studies
13	Mustafa et al.	2022	Measured universities: case studies
14	Cano et al.	2022	Measured universities: case studies
15	Osorio et al.	2022	Measured universities: case studies
16	Battistini et al.	2023	Measured universities: case studies
17	Ahonen et al.	2024	Measured universities: case studies
18	Rus et al.	2025	Measured universities: case studies
19	Yang et al.	2025	Measured universities: case studies
20	Wagner et al.	2021	Schools
21	Adhikari and Pokhrel	2025	Schools
22	Badr, El Beheary and Hassan	2026	Egypt and public buildings
23	Safwat et al.	2024	Egypt and public buildings
24	Hamada, Hong and Raslan	2025	Egypt and public buildings

#	Paper	Year	Group
25	García-Monge et al.	2023	Egypt and public buildings
26	Akenji et al.	2021	Targets and national data
27	Crippa et al. (EDGAR)	2024	Targets and national data
28	Ajufo and Bekaroo	2021	Technology for the app and the monitor
29	Bekaroo et al.	2021	Technology for the app and the monitor
30	Zhang et al.	2026	Technology for the app and the monitor
31	Sadeghian, Håkansson and Zhao	2021	Technology for the app and the monitor
32	Wang et al. (SHL)	2021	Technology for the app and the monitor
33	Wang and Jiang	2022	Technology for the app and the monitor
34	Tahir and Loo	2021	Technology for the app and the monitor
35	Dalakleidi et al.	2022	Technology for the app and the monitor
36	Ismail and Sahari	2024	Technology for the app and the monitor
37	Miller et al.	2020	Technology for the app and the monitor
38	Anand et al.	2021	Technology for the app and the monitor
39	Himeur et al.	2021	Technology for the app and the monitor
40	Azad, Rajabi and Estebsari	2023	Technology for the app and the monitor

12.1 How an educational institution's footprint is measured

Almost every study uses the three scopes of the GHG Protocol Corporate Standard [1, 10, 14, 18, 22], and some also follow ISO 14064-1 and ISO/TR 14069 [14, 15, 16]:

Scope	Examples in a university or school	How Basmati measures it
Scope 1 · direct	Gas for heating and kitchens, diesel generators, the institution's buses, refrigerant leaks	Optional manual entry
Scope 2 · purchased electricity	Labs, PCs, air conditioning and lighting	Measured every 10 s by the floor monitors
Scope 3 · value chain	Student and staff commuting, food, purchases, waste, water, business travel	Commuting and meals per person, from the app

- **Steps:** set the boundary and base year (a whole campus, a building [4] or one faculty [22]); list the sources per scope; collect activity data (bills and meters, commuting surveys, purchase records); choose local emission factors; multiply activity by factor; normalize per person or per m² [3, 5, 18]; check uncertainty and verify [15, 19]; report and set reduction targets [17].
- **Methods:** activity-based calculation (measured quantities × factors) is the most transparent [10, 14, 18, 19]; spending-based input-output estimates cover purchases but are weak on their own [6]; hybrid models combine both [7]; life cycle assessment is the most complete [11].
- **Schools:** the same GHG Protocol with simpler data, from student-led tools [20] to questionnaires and school records [21].

12.2 What measured campuses show

Institution	Year	t CO_2e	Per person	Largest source	Ref.
Clemson University, USA	2014–17	≈ 95,400	—	Electricity and Scope 3, ≈ 40% each	[11]
Univ. Nacional de Colombia	2019	7,251	0.43 t	Transport 58.5%; Scope 3 83%	[14]
NED University, Pakistan	2017	≈ 21,500	1.79 t	Scope 3 85.6%	[13]
TU Delft, Netherlands	2018	≈ 106,000	—	Scope 3 ≈ 80% (purchases)	[6]
University of Oulu, Finland	2019	19,072	—	District heating	[7]
University of Bologna, Italy	2020	16,467	—	IT purchases ≈ 50% (transport 74% in 2018)	[16]
TU Cluj-Napoca, Romania	2023	7,229	0.36 t	Natural gas for heating	[18]
Sichuan Univ. Jinjiang College, China	2023	44,584	1.89 t	Electricity	[19]
Port Said University, Egypt	2023–24	—	—	Scope 3 (commuting) ≈ 97%	[22]
8 schools in Pokhara, Nepal	—	409.8 (all)	0.029 t	School buses and fuels (Scope 1)	[21]

- Scope 3, mainly commuting, is usually the largest part (80–97%) [6, 13, 14, 22], and it is the most uncertain item [19]; so the app measures every volunteer's commute automatically.
- Where commuting is not counted or heating is by gas, building energy dominates [7, 18, 19]; in Egyptian public buildings about 40% of electricity is cooling [24].
- Boundaries, methods and years differ, so totals are not directly comparable [1, 3, 5]; we report per person and per person-hour with one fixed method.
- The mix changes quickly when presence changes, as during COVID-19 [12, 16]; this supports daily rather than yearly measurement.
- Institutional reports also exist in the region: the American University in Cairo since 2012, Qatar University since 2016, and Alexandria University in its UI GreenMetric submissions.

12.3 Measuring educational institutions: reviews, methods and tools

[1] Valls-Val and Bovea (2021)

Valls-Val, K., & Bovea, M. D. (2021). Carbon footprint in higher education institutions: a literature review and prospects for future research. Clean Technologies and Environmental Policy, 23, 2523–2542. doi:10.1007/s10098-021-02180-2

Summary: Systematic review of how universities assess their carbon footprint (studies up to March 2021): scopes, methods and emission sources, with directions for future research.

What we take: Frames our floor as a university facility and shows that assessments are usually yearly and inconsistent.

Link: https://doi.org/10.1007/s10098-021-02180-2 · Repository copy: repositori.uji.es (handle 10234/195375)

[2] Li et al. (2021)

Li, Z., Chen, Z., Yang, N., Wei, K., Ling, Z., Liu, Q., Chen, G., & Ye, B. H. (2021). Trends in research on the carbon footprint of higher education: a bibliometric analysis (2010–2019). Journal of Cleaner Production, 289, 125642. doi:10.1016/j.jclepro.2020.125642

Summary: Bibliometric analysis of 283 papers (2010–2019, Web of Science) on the carbon footprint of higher education: who publishes, where, and which topics grow; the University of California system and the USA lead, and the Journal of Cleaner Production is the main outlet.

What we take: Maps the field and shows that measuring campus footprints is an established research area we can build on.

Link: https://doi.org/10.1016/j.jclepro.2020.125642 · Elsevier (check the university library)

[3] Helmers, Chang and Dauwels (2021)

Helmers, E., Chang, C. C., & Dauwels, J. (2021). Carbon footprinting of universities worldwide: Part I—objective comparison by standardized metrics. Environmental Sciences Europe, 33, 30. doi:10.1186/s12302-021-00454-6

Summary: Collects carbon footprint data from 20 universities worldwide; reported footprints are hard to compare because methods, measures and targets differ; proposes standardized metrics.

What we take: Supports reporting per person and per person-hour (our intensity I), not only totals.

Link: https://doi.org/10.1186/s12302-021-00454-6 · Open access

[4] Valls-Val and Bovea (CO2UNV) (2022)

Valls-Val, K., & Bovea, M. D. (2022). Carbon footprint assessment tool for universities: CO2UNV. Sustainable Production and Consumption, 29, 791–804. doi:10.1016/j.spc.2021.11.020

Summary: CO2UNV, a calculation tool that estimates a university's emissions in Scopes 1, 2 and 3, for the whole university or for single buildings, with default emission sources and factors for educational centres, tracking over time and offsets; applied and validated at Universitat Jaume I.

What we take: Shows that a footprint can be computed per building, which is exactly our per-floor approach; its list of emission sources guides our scope coverage.

Link: https://doi.org/10.1016/j.spc.2021.11.020 · Repository copy: repositori.uji.es (handle 10234/196551)

[5] Guerrero-Lucendo et al. (2022)

Guerrero-Lucendo, A., García-Orenes, F., Navarro-Pedreño, J., & Alba-Hidalgo, D. (2022). General mapping of the environmental performance in climate change mitigation of Spanish universities through a standardized carbon footprint calculation tool. International Journal of Environmental Research and Public Health, 19(17), 10964. doi:10.3390/ijerph191710964

Summary: Compares the climate performance of Spanish universities by applying one standardized carbon footprint calculation tool to all of them, so that results become comparable.

What we take: Supports using one fixed method and one set of factors for every floor and person, so results can be compared over time.

Link: https://doi.org/10.3390/ijerph191710964 · Open access

[6] Herth and Blok (2023)

Herth, A., & Blok, K. (2023). Quantifying universities' direct and indirect carbon emissions: the case of Delft University of Technology. International Journal of Sustainability in Higher Education, 24(9). doi:10.1108/IJSHE-04-2022-0121

Summary: Calculates TU Delft's 2018 footprint (utilities, logistics and purchases) with process analysis and economic input-output analysis: about 106 kt CO_2e, 80% of it Scope 3; argues that input-output models alone are a questionable way to calculate a university's footprint.

What we take: Supports measuring activities directly (kWh, km, meals) instead of estimating from spending, which is what our monitors and app do.

Link: https://doi.org/10.1108/IJSHE-04-2022-0121 · Open access (Emerald)

[7] Kiehle et al. (2023)

Kiehle, J., Kopsakangas-Savolainen, M., Hilli, M., & Pongrácz, E. (2023). Carbon footprint at institutions of higher education: the case of the University of Oulu. Journal of Environmental Management, 329, 117056. doi:10.1016/j.jenvman.2022.117056

Summary: Footprint of the University of Oulu with a hybrid model combining environmentally extended input-output analysis and life cycle assessment: 19,072 t CO_2e in 2019, with district heating on campus as the largest share.

What we take: A complete, recent method for one university; shows that building energy (heat in Finland, cooling in Egypt) is often the biggest item.

Link: https://doi.org/10.1016/j.jenvman.2022.117056 · Open access (CC BY)

[8] Deda, Gervásio and Quina (2025)

Deda, D., Gervásio, H., & Quina, M. J. (2025). Advancing carbon footprint in higher education: an integrated assessment model. The International Journal of Life Cycle Assessment. doi:10.1007/s11367-025-02514-y

Summary: Proposes an integrated model for assessing the carbon footprint of higher education institutions, developed at the University of Coimbra.

What we take: The most recent methodological proposal (2025); useful for the methods chapter of our report.

Link: https://doi.org/10.1007/s11367-025-02514-y · Springer (check the university library)

[9] Abdelbasier et al. (2022)

Abdelbasier, A. M., Kamel, M. M., Donia, N. S., & Nawar, H. A. (2022). Carbon footprint monitoring and mitigation in higher education: a meta-analysis review. Journal of Environmental Sciences (Ain Shams University), 51(11).

Summary: Egyptian review (Ain Shams University journal) that compares carbon footprint monitoring and reduction methods at four campuses (BITS Pilani, AUS Sharjah, King's College London, St. Edward's University); electricity is a major share of campus emissions.

What we take: An Egyptian source that confirms electricity as a main campus source, which our floor monitors measure.

Link: https://jes.journals.ekb.eg/article_295030.html · Open on the Egyptian Knowledge Bank (EKB)

12.4 Measured universities: case studies

[10] Yañez, Sinha and Vásquez (2020)

Yañez, P., Sinha, A., & Vásquez, M. (2020). Carbon footprint estimation in a university campus: evaluation and insights. Sustainability, 12(1), 181. doi:10.3390/su12010181

Summary: University of Talca (Chile), which has measured its footprint every year since 2012 with the GHG Protocol across five campuses; explains how a campus footprint is determined and which factors drive it; Scopes 1 and 2 are about 40% of the Talca campus total.

What we take: A clear step-by-step GHG Protocol example for a campus, and a model for yearly tracking.

Link: https://doi.org/10.3390/su12010181 · Open access

[11] Clabeaux et al. (2020)

Clabeaux, R., Carbajales-Dale, M., Ladner, D., & Walker, T. (2020). Assessing the carbon footprint of a university campus using a life cycle assessment approach. Journal of Cleaner Production, 273, 122600. doi:10.1016/j.jclepro.2020.122600

Summary: Streamlined life cycle assessment of Clemson University (USA) as a baseline for mitigation, covering steam, refrigerants, electricity, transport, wastewater and paper; about 95,400 t CO_2e, split roughly 19% Scope 1, 41% Scope 2 and 40% Scope 3.

What we take: Shows the full list of campus emission sources and that purchased electricity can be as large as all of Scope 3.

Link: https://cecas.clemson.edu/ladnergroup/blog/wp-content/uploads/2020/07/Clabeaux-et-al-2020-CO2-Footprint-Journal-Cleaner-Production.pdf · Free accepted manuscript (Clemson)

[12] Filimonau et al. (2021)

Filimonau, V., Archer, D., Bellamy, L., Smith, N., & Wintrip, R. (2021). The carbon footprint of a UK University during the COVID-19 lockdown. Science of the Total Environment, 756, 143964. doi:10.1016/j.scitotenv.2020.143964

Summary: Bournemouth University's footprint fell by almost 30% during the April–June 2020 lockdown, but the carbon intensity of online teaching was almost equal to that of staff and student commuting before the lockdown.

What we take: Shows that commuting is a major item and that moving online is not automatically low-carbon; supports measuring both presence and commuting.

Link: https://eprints.bournemouth.ac.uk/34947/ · Free accepted manuscript (Bournemouth repository)

[13] Mustafa et al. (2022)

Mustafa, A., Kazmi, M., Khan, H. R., Qazi, S. A., & Lodi, S. H. (2022). Towards a carbon neutral and sustainable campus: case study of NED University of Engineering and Technology. Sustainability, 14(2), 794. doi:10.3390/su14020794

Summary: NED University (Karachi, Pakistan), 2017: about 21,500 t CO_2e, or 1.79 t per student; Scopes 1 and 2 about 7% each and Scope 3 85.6%; recommends renewables, efficient appliances, electric vehicles and tree planting.

What we take: A developing-country campus similar to Egyptian ones, where commuting-related Scope 3 dominates.

Link: https://doi.org/10.3390/su14020794 · Open access

[14] Cano et al. (2022)

Cano, N., Berrio, L., Carvajal, E., & Arango, S. (2022). Assessing the carbon footprint of a Colombian university campus using the UNE-ISO 14064–1 and WRI/WBCSD GHG Protocol Corporate Standard. Environmental Science and Pollution Research. doi:10.1007/s11356-022-22119-4

Summary: Universidad Nacional de Colombia, Medellín, 2019, measured with UNE-ISO 14064-1 and the GHG Protocol: about 7,251 t CO_2e, 0.43 t per person; Scope 3 83% and transport alone 58.5%.

What we take: The clearest example of applying ISO 14064-1 and the GHG Protocol to a campus; transport is the largest item, as we expect in Egypt.

Link: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9374572/ · Open access (PubMed Central)

[15] Osorio et al. (2022)

Osorio, A. M., Úsuga, L. F., Vásquez, R. E., Nieto-Londoño, C., Rinaudo, M. E., Martínez, J. A., & Leal Filho, W. (2022). Towards carbon neutrality in higher education institutions: case of two private universities in Colombia. Sustainability, 14(3), 1774. doi:10.3390/su14031774

Summary: How two Colombian universities became certified carbon neutral (the first in Latin America in 2017): GHG inventory under ISO 14064 in three phases, scoping and quantification, analysis and mitigation, then verification and offsetting.

What we take: Shows the full path from measuring to reducing to verified neutrality, the long-term goal our measurements serve.

Link: https://doi.org/10.3390/su14031774 · Open access

[16] Battistini et al. (2023)

Battistini, R., Passarini, F., Marrollo, R., Lantieri, C., Simone, A., & Vignali, V. (2023). How to assess the carbon footprint of a large university? The case study of University of Bologna's multicampus organization. Energies, 16(1), 166. doi:10.3390/en16010166

Summary: University of Bologna, measured with the GHG Protocol, ISO 14064 and ISO/TR 14069: 16,467 t CO_2e in 2020 (15,753 after on-site renewables) versus 55,388 t in 2018; transport was 74% in 2018, while IT purchases were almost half in 2020.

What we take: Shows how much the mix changes when people stay home, so daily measurement of presence and commuting matters.

Link: https://doi.org/10.3390/en16010166 · Open access

[17] Ahonen et al. (2024)

Ahonen, V. L., Woszczek, A., Baumeister, S., Helimo, U. T., Jackson, A. K., Kopsakangas-Savolainen, M., Kääriä, J., Lehtonen, T., Luoranen, M., Pongrácz, E., Soukka, R., Vainio, V., & El Geneidy, S. (2024). Carbon neutral higher education institutions: a reality check, challenges and solutions. International Journal of Sustainability in Higher Education, 25(9), 293–315. doi:10.1108/IJSHE-11-2023-0515

Summary: Survey of sustainability experts at all 38 Finnish higher education institutions with a SWOT analysis: most track their carbon footprint, but only a minority have separate reduction targets and clear offsetting plans; smaller institutions lack resources.

What we take: Measuring alone is not enough; our dashboard links measurement to targets (the floor baseline) and actions (waste alerts).

Link: https://doi.org/10.1108/IJSHE-11-2023-0515 · Open access (CC BY)

[18] Rus et al. (2025)

Rus, T., Moldovan, R.-P., Beu, D., Pop, M., & Ceclan, A. (2025). Towards a climate-neutral campus: carbon footprint assessment in higher education institutions. Applied Sciences, 15(7), 3695. doi:10.3390/app15073695

Summary: Technical University of Cluj-Napoca (Romania), all three scopes in 2022–2023 with the GHG Protocol and One Click LCA: 7,445 and 7,229 t CO_2e; 0.378 then 0.362 t per person; natural gas for heating is the largest source.

What we take: A recent two-year inventory with per-person and per-m^2 indicators, the same kind of intensity we report for the floor.

Link: https://doi.org/10.3390/app15073695 · Open access

[19] Yang et al. (2025)

Yang, C., Yao, T., Dong, S., & Jiang, W. (2025). Carbon emissions accounting and uncertainty analysis in campus settings: a case study of a university in Sichuan, China. PLOS ONE. doi:10.1371/journal.pone.0321216

Summary: Sichuan University Jinjiang College, 2023, with the emission factor method and a Monte Carlo uncertainty analysis: net 44,584 t CO_2e, 1.89 t per person; electricity is the largest source, and commuting and business travel carry the largest uncertainty.

What we take: Shows that commuting is the most uncertain part of a campus inventory; measuring trips automatically, as our app does, attacks that uncertainty directly.

Link: https://doi.org/10.1371/journal.pone.0321216 · Open access

12.5 Schools

[20] Wagner et al. (2021)

Wagner, O., Tholen, L., Nawothnig, L., & Albert-Seifried, S. (2021). Making school-based GHG-emissions tangible by student-led carbon footprint assessment program. Energies, 14(24), 8558. doi:10.3390/en14248558

Summary: Schools4Future (Germany): a tool that students use to calculate their school's footprint from mobility, heating, electricity, canteen food and paper, tested in pilot schools; commuting was about 60–70 t CO_2 per school, and the canteen was the smallest share.

What we take: A model for schools and for involving students in measurement, like our volunteer mode.

Link: https://doi.org/10.3390/en14248558 · Open access

[21] Adhikari and Pokhrel (2025)

Adhikari and Pokhrel (2025). Study on the carbon footprint of schools in Pokhara, Nepal. Himalayan Journal of Applied Science and Engineering, 5(2), 173–183. doi:10.3126/hijase.v5i2.74787

Summary: Eight public and private schools (each with over 1,000 students) measured with the GHG Protocol using questionnaires, interviews and records: 409.8 t CO_2e per year in total, 0.029 t per person; Scope 1 (school buses and fuels) is the largest, then Scope 3.

What we take: Shows how to measure schools with simple data in a developing country, and that school transport dominates.

Link: https://www.nepjol.info/index.php/hijase/article/view/74787 · Open access (NepJOL)

12.6 Egypt, public buildings and continuous measurement

[22] Badr, El Beheary and Hassan (2026)

Badr, M. K., El Beheary, M., & Hassan, I. A. (2026). The importance of calculating the carbon footprint of Egyptian universities and their role in sustainable development: Port Said University as case study. Annals of the Academy of Romanian Scientists, Series on Biological Sciences. doi:10.56082/annalsarscibio.2026.1.191

Summary: Port Said University (Egypt): pilot inventory of the Faculty of Science campus from 2023 with the GHG Protocol; Scope 3, mainly commuting, was about 97% of emissions in 2023 and 2024; Scopes 1 and 2 fell in 2024 after resource-management and clean-transport measures.

What we take: The closest study to ours: an Egyptian government university, measured per faculty, with commuting as the dominant source.

Link: https://doi.org/10.56082/annalsarscibio.2026.1.191 · Journal site (aos.ro)

[23] Safwat et al. (2024)

Safwat, H., Abdel-Rehim, A. A., El-Mahallawi, I., Hussein, A. A., Amer, A. M., Elshazly, E., & Elshamy, A. I. (2024). Guidelines for data collection on energy performance of higher-education buildings in Egypt: a case study. HBRC Journal, 20(1), 615–642. doi:10.1080/16874048.2024.2345553

Summary: Egyptian university buildings usually lack monthly electricity and gas records per building, so the authors build equipment and HVAC datasheets to estimate consumption; in their case study, PCs, laboratories, lighting and air conditioning drive electricity use.

What we take: Egyptian evidence that the data our monitors produce does not exist today; we reuse their survey idea in week 2.

Link: https://doi.org/10.1080/16874048.2024.2345553 · Open access (also openresearch.lsbu.ac.uk/item/97z76)

[24] Hamada, Hong and Raslan (2025)

Hamada, A. A., Hong, S.-M., & Raslan, R. (2025). Evaluating the operational performance in Cairo's public administrative buildings: stock characterisation and typical energy use. Energy and Buildings, 346, 116065. doi:10.1016/j.enbuild.2025.116065

Summary: First characterisation of Cairo's public administrative buildings from metered data: about 40% of electricity goes to cooling and 60% to lighting and equipment; a winter month uses about 5% of the annual energy and a summer peak month about 13%.

What we take: Shows where a government building's energy goes and why temperature is a forecast input.

Link: https://discovery.ucl.ac.uk/id/eprint/10211499 · Open access copy at UCL Discovery

[25] García-Monge et al. (2023)

García-Monge, M., Zalba, B., Casas, R., Cano, E., Guillén-Lambea, S., López-Mesa, B., & Martínez, I. (2023). Is IoT monitoring key to improve building energy efficiency? Case study of a smart campus in Spain. Energy and Buildings, 285, 112882. doi:10.1016/j.enbuild.2023.112882

Summary: Continuous IoT monitoring on a university campus in Zaragoza, used to support energy decisions; the authors report that the approach can save 40–70% of HVAC energy, and stress CO_2 monitoring in rooms with changing occupancy.

What we take: Direct evidence that measuring a campus building continuously pays off, especially for air conditioning.

Link: https://doi.org/10.1016/j.enbuild.2023.112882 · Repository copy: zaguan.unizar.es/record/130169

12.7 Targets and national data

[26] Akenji et al. (2021)

Akenji, L., Bengtsson, M., Toivio, V., Lettenmeier, M., et al. (2021). 1.5-Degree Lifestyles: Towards a fair consumption space for all. Hot or Cool Institute, Berlin.

Summary: Sets per-person targets for lifestyle carbon footprints compatible with 1.5 °C: about 2.5 t CO_2e by 2030, falling to 0.7 t by 2050, and compares the lifestyle footprints of ten countries.

What we take: The 2.5 t target gives our personal daily budget: 2,500 ÷ 365 ≈ 6.8 kg CO_2e.

Link: https://hotorcool.org/1-5-degree-lifestyles-report · Free download from the institute

[27] Crippa et al. (EDGAR) (2024)

Crippa, M., Guizzardi, D., Pagani, F., et al. (2024). GHG emissions of all world countries: 2024 report. Publications Office of the European Union, Luxembourg. JRC138862. doi:10.2760/4002897

Summary: Greenhouse gas emissions of every country from 1970 to 2023, by sector, from the EDGAR database of the European Commission's Joint Research Centre and the IEA.

What we take: Source of the per-person figures for Egypt (3.4 t) and the world (6.6 t) in 2023.

Link: https://data.europa.eu/doi/10.2760/4002897 · Free (EU Publications Office / edgar.jrc.ec.europa.eu)

12.8 Technology for the app and the monitor

[28] Ajufo and Bekaroo (2021)

Ajufo, C., & Bekaroo, G. (2021). An automated personal carbon footprint calculator for estimating carbon emissions from transportation use. In Proceedings of the International Conference on Artificial Intelligence and its Applications (icARTi '21). ACM. doi:10.1145/3487923.3487935

Summary: TCTracker, a mobile app that estimates a person's transport emissions automatically from GPS with AI-based mode detection; evaluated with 40 users using the Technology Acceptance Model.

What we take: Closest earlier app to ours; shows automatic tracking is accepted by users. We add food, a building share and Egyptian factors.

Link: https://doi.org/10.1145/3487923.3487935 · ACM Digital Library (check the university library)

[29] Bekaroo et al. (2021)

Bekaroo, G., Roopowa, D., Zakari, A., & Niemeier, D. (2021). Calculating carbon emissions from personal travelling: insights from a top-down analysis of key calculators. Environmental Science and Pollution Research, 28(7), 8853–8872. doi:10.1007/s11356-020-11179-z

Summary: Analyses ten transport carbon calculators: they differ in granularity, have limited transparency, sometimes give very different results for the same trip, and do not fully engage users.

What we take: Motivates an automatic, transparent calculator with documented factors.

Link: https://doi.org/10.1007/s11356-020-11179-z · Free author copy: repository.mdx.ac.uk/item/8929y

[30] Zhang et al. (2026)

Zhang, Y., Echchabi, O., Feng, T., Zhang, W., Liao, H.-K., & Chang, C. (2026). Detecting transportation mode using dense smartphone GPS trajectories and Transformer models. International Journal of Geographical Information Science. arXiv:2603.00340

Summary: SpeedTransformer, a Transformer model that detects transport mode from dense phone GPS speed data; tested in the CarbonClever app with 348 users to estimate personal transport footprints.

What we take: The most recent system like ours; confirms that automatic mode detection works for personal footprints at the scale of hundreds of users.

Link: https://arxiv.org/abs/2603.00340 · Free on arXiv

[31] Sadeghian, Håkansson and Zhao (2021)

Sadeghian, P., Håkansson, J., & Zhao, X. (2021). Review and evaluation of methods in transport mode detection based on GPS tracking data. Journal of Traffic and Transportation Engineering (English Edition), 8(4), 467–482.

Summary: Reviews and compares transport mode detection methods on GPS data. Most are supervised and need manually labeled trips, which is slow and error-prone; deep learning is promising for large unlabeled data.

What we take: Labeled local trips are the bottleneck, hence our volunteer labeling mode and one-tap confirmation.

Link: https://doaj.org/article/cedcdc9d16f541aaa07a5c67ef7f5d2b · Open access

[32] Wang et al. (SHL) (2021)

Wang, L., Gjoreski, H., Ciliberto, M., Lago, P., Murao, K., Okita, T., & Roggen, D. (2021). Three-year review of the 2018–2020 SHL challenge on transportation and locomotion mode recognition from mobile sensors. Frontiers in Computer Science, 3, 713719. doi:10.3389/fcomp.2021.713719

Summary: Reviews three years of the SHL challenge (8 modes from phone motion sensors, 48 submissions). Most modes are recognized well, but separating rail from road transport remains hard.

What we take: Our metro rule (GPS loss underground plus station geofences) targets exactly the rail-versus-road confusion.

Link: https://doi.org/10.3389/fcomp.2021.713719 · Open access

[33] Wang and Jiang (2022)

Wang, P., & Jiang, Y. (2022). Transportation mode detection using temporal convolutional networks based on sensors integrated into smartphones. Sensors, 22(17), 6712. doi:10.3390/s22176712

Summary: T2Trans, a temporal convolutional network on lightweight phone sensors; macro F1 of 86.42% on SHL and 88.37% on the HTC dataset for 8 modes.

What we take: A realistic reference for our 85% transport target.

Link: https://doi.org/10.3390/s22176712 · Open access

[34] Tahir and Loo (2021)

Tahir, G. A., & Loo, C. K. (2021). A comprehensive survey of image-based food recognition and volume estimation methods for dietary assessment. Healthcare, 9(12), 1676. doi:10.3390/healthcare9121676

Summary: Surveys food recognition and volume estimation methods and the mobile apps that use them; about two thirds of the studies use deep visual features, and all use CNN variants for ingredient recognition.

What we take: Supports transfer learning on a lightweight CNN, and a user-chosen portion size instead of volume estimation.

Link: https://doi.org/10.3390/healthcare9121676 · Open access (also arXiv:2106.11776)

[35] Dalakleidi et al. (2022)

Dalakleidi, K. V., Papadelli, M., Kapolos, I., & Papadimitriou, K. (2022). Applying image-based food-recognition systems on dietary assessment: a systematic review. Advances in Nutrition, 13(6), 2590–2619. doi:10.1093/advances/nmac078

Summary: Systematic review of image-based food recognition systems used for dietary assessment; 78 studies reviewed in full text.

What we take: Shows how such systems are evaluated in practice; we report Top-1 and Top-3 accuracy on new people.

Link: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9776640/ · Open access (PubMed Central)

[36] Ismail and Sahari (2024)

Ismail, M. L., & Sahari, N. (2024). Real time energy monitoring & current spikes detection system for enhanced energy conservation in residential area. Progress in Engineering Application and Technology, 5(1), 79–87. doi:10.30880/peat.2024.05.01.008

Summary: ESP32 with a ZMPT101B voltage sensor and an SCT-013-030 current clamp, sending data to ThingSpeak and a mobile app; tracks voltage, current, energy and bill, and flags current spikes. Worst-case error about 12.4%.

What we take: Same board and clamp family as our monitor; its error is why we calibrate weekly against a reference.

Link: https://penerbit.uthm.edu.my/periodicals/index.php/peat/article/view/14767 · Open access

[37] Miller et al. (2020)

Miller, C., et al. (2020). The ASHRAE Great Energy Predictor III competition: overview and results. Science and Technology for the Built Environment, 26(10). doi:10.1080/23744731.2020.1795514

Summary: Overview of a large competition on forecasting hourly energy meters of non-residential buildings; the winning solutions relied on gradient boosting with careful preprocessing.

What we take: Why the floor forecast starts with LightGBM; its dataset (Building Data Genome 2) is our public benchmark.

Link: https://arxiv.org/abs/2007.06933 · Free on arXiv

[38] Anand et al. (2021)

Anand, P., Deb, C., Yang, K., Yang, J., Cheong, D., & Sekhar, C. (2021). Occupancy-based energy consumption modelling using machine learning algorithms for institutional buildings. Energy and Buildings, 111478. doi:10.1016/j.enbuild.2021.111478

Summary: Models energy from occupancy for eight space types in three institutional buildings; the occupancy–energy link differs by space and time; a deep network is slightly more accurate, gradient boosting much faster.

What we take: Justifies the timetable (occupancy) as a forecast input and gradient boosting as the first model.

Link: https://doi.org/10.1016/j.enbuild.2021.111478 · Repository record: research-collection.ethz.ch (handle 20.500.11850/508529)

[39] Himeur et al. (2021)

Himeur, Y., Ghanem, K., Alsalemi, A., Bensaali, F., & Amira, A. (2021). Artificial intelligence based anomaly detection of energy consumption in buildings: a review, current trends and new perspectives. Applied Energy, 287, 116601. doi:10.1016/j.apenergy.2021.116601

Summary: Review and taxonomy of AI methods for detecting abnormal energy use in buildings, by algorithm, features, detection level, computing platform and application.

What we take: Guides our waste detector: simple rules first, then an unsupervised model on forecast errors.

Link: https://arxiv.org/abs/2010.04560 · Free preprint on arXiv

[40] Azad, Rajabi and Estebsari (2023)

Azad, Rajabi and Estebsari (2023). Non-intrusive load monitoring (NILM) using deep neural networks: a review. arXiv:2306.05017

Summary: Review of deep neural network approaches to non-intrusive load monitoring and the public datasets used to train them.

What we take: Model and dataset choices for the optional NILM step.

Link: https://arxiv.org/abs/2306.05017 · Free on arXiv

12.9 Research gap and our contribution

Campuses measure their footprint once a year, for the whole campus, from bills and commuting surveys [1, 10, 14, 18, 22]; personal calculators stop at the person [29]. Nothing measures a public building and its people every day.

- **Automatic and daily:** commute, meals and a share of the floor's electricity combined into one number, with almost no typing, instead of a yearly commuting survey [19, 28, 30].
- **Local factors and data:** microbus, metro, Egyptian dishes and the Egyptian grid, with data we collect ourselves [31–35].
- **Measured public building:** the floor's real electricity, each person's share and after-hours waste for the facility manager, where Egyptian buildings lack data today [22–25, 39].

13. Plan

13.1 Timeline

Weeks	Work	Milestone
1–2	Interfaces (MQTT, API, model inputs); floor survey; official approval request; order parts	Week 2: interfaces agreed
3–5	App screens, database, trip tracking prototype; dish list and recipes; monitor tested on a kettle in the lab	—
6	Monitors installed on the floor board by the faculty electrician; volunteer mode live with 20 people	Week 6: monitors and volunteers
7–10	Data collection; first models; daily engine end to end; first curves	Week 10: first full version
11–12	Mid-year exams: light work; monitors keep recording	—
13–16	Improve models; on-phone inference; forecast, waste detection and dashboard	Week 16: feature freeze
17–18	Final evaluation on held-out people and weeks; battery tests	—
19–20	Report, demo and defense	Week 20: final delivery

13.2 Team

Team	People	Owns
Mobile app	3	Trip tracking; food camera and on-phone models; screens, check-in and volunteer mode
AI	2	Meal model and dish database; transport model; waste detection
Backend	2	API, database, daily calculation, MQTT, forecast, dashboard
Hardware	2	Circuits and enclosures; installation with the faculty electrician; calibration; entrance display

Extra roles inside the teams: a team lead, a volunteer coordinator and a report editor.

13.3 Budget (estimates)

Item	Details	EGP
Floor monitors	3 ESP32 boards, 7 CT clamps (six plus a spare), 3 ADS1115, DIN-rail enclosures, burden resistors, 5 V adapters	5,500
Entrance display	LED ring in a small enclosure	600
Printing	Report and poster	1,000
Kitchen scale	Weighing dish ingredients	300
Contingency	10%	740
Server	Faculty server or a free cloud tier	0
Total		**8,140**

Prices are estimates at the time of writing. The reference power meter is borrowed from the electrical lab.

13.4 Risks

Risk	Mitigation
Installation approval is delayed	Request it in week 1; meanwhile test on a lab circuit and pretrain the forecast on Building Data Genome 2
Main-phase current is above the clamp range, or the board is hard to reach	100 A clamps on main phases; electrician survey in week 2; measure sub-circuits if needed
Few volunteers or low engagement	Recruit from department courses; one tap per trip; weekly feedback
Meal model weak on Egyptian dishes	200+ photos per dish; Top-3 with user confirmation
Monitor error	Weekly calibration factor k against a reference reading
Privacy concerns	On-phone processing, IDs, consent, hours-only presence, raw GPS deleted

14. Next steps

Decisions we need from the supervisor

- Approval of the idea and the pilot on the CS floor
- An official letter to the faculty administration for installation
- Access to the timetable and the building’s past electricity bills
- Consent form for 20 volunteers
- Final team names and roles

Our next two weeks

- Interfaces: MQTT, API and model inputs
- Floor survey: rooms, labs, ACs, PCs and the electrical board
- Order parts; test one clamp on a kettle in the lab
- Dish list with the cafeteria menu; consent form

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Appendix A. Where to download each paper

Most papers are open access. For the others, a free author copy or preprint is listed; otherwise use the university library login.

#	Paper	Link	Access
1	Valls-Val and Bovea (2021)	doi.org/10.1007/s10098-021-02180-2	Repository copy: repositori.uji.es (handle 10234/195375)
2	Li et al. (2021)	doi.org/10.1016/j.jclepro.2020.125642	Elsevier (check the university library)
3	Helmers, Chang and Dauwels (2021)	doi.org/10.1186/s12302-021-00454-6	Open access
4	Valls-Val and Bovea (CO2UNV) (2022)	doi.org/10.1016/j.spc.2021.11.020	Repository copy: repositori.uji.es (handle 10234/196551)
5	Guerrero-Lucendo et al. (2022)	doi.org/10.3390/ijerph191710964	Open access
6	Herth and Blok (2023)	doi.org/10.1108/IJSHE-04-2022-0121	Open access (Emerald)
7	Kiehle et al. (2023)	doi.org/10.1016/j.jenvman.2022.117056	Open access (CC BY)
8	Deda, Gervásio and Quina (2025)	doi.org/10.1007/s11367-025-02514-y	Springer (check the university library)
9	Abdelbasier et al. (2022)	jes.journals.ekb.eg/article_295030.html	Open on the Egyptian Knowledge Bank (EKB)
10	Yañez, Sinha and Vásquez (2020)	doi.org/10.3390/su12010181	Open access
11	Clabeaux et al. (2020)	cecas.clemson.edu/ladnergroup/blog/wp-content/uploads/2020/07/Clabeaux-et-al-2020-CO2-Footprint-Journal-Cleaner-Production.pdf	Free accepted manuscript (Clemson)
12	Filimonau et al. (2021)	eprints.bournemouth.ac.uk/34947/	Free accepted manuscript (Bournemouth repository)
13	Mustafa et al. (2022)	doi.org/10.3390/su14020794	Open access
14	Cano et al. (2022)	www.ncbi.nlm.nih.gov/pmc/articles/PMC9374572/	Open access (PubMed Central)
15	Osorio et al. (2022)	doi.org/10.3390/su14031774	Open access
16	Battistini et al. (2023)	doi.org/10.3390/en16010166	Open access
17	Ahonen et al. (2024)	doi.org/10.1108/IJSHE-11-2023-0515	Open access (CC BY)
18	Rus et al. (2025)	doi.org/10.3390/app15073695	Open access
19	Yang et al. (2025)	doi.org/10.1371/journal.pone.0321216	Open access
20	Wagner et al. (2021)	doi.org/10.3390/en14248558	Open access
21	Adhikari and Pokhrel (2025)	www.nepjol.info/index.php/hijase/article/view/74787	Open access (NepJOL)

#	Paper	Link	Access
22	Badr, El Beheary and Hassan (2026)	doi.org/10.56082/annalsarscibio.2026.1.191	Journal site (aos.ro)
23	Safwat et al. (2024)	doi.org/10.1080/16874048.2024.2345553	Open access (also openresearch.lsbu.ac.uk/item/97z76)
24	Hamada, Hong and Raslan (2025)	discovery.ucl.ac.uk/id/eprint/10211499	Open access copy at UCL Discovery
25	García-Monge et al. (2023)	doi.org/10.1016/j.enbuild.2023.112882	Repository copy: zaguan.unizar.es/record/130169
26	Akenji et al. (2021)	hotorcool.org/1-5-degree-lifestyles-report	Free download from the institute
27	Crippa et al. (EDGAR) (2024)	data.europa.eu/doi/10.2760/4002897	Free (EU Publications Office / edgar.jrc.ec.europa.eu)
28	Ajufo and Bekaroo (2021)	doi.org/10.1145/3487923.3487935	ACM Digital Library (check the university library)
29	Bekaroo et al. (2021)	doi.org/10.1007/s11356-020-11179-z	Free author copy: repository.mdx.ac.uk/item/8929y
30	Zhang et al. (2026)	arxiv.org/abs/2603.00340	Free on arXiv
31	Sadeghian, Håkansson and Zhao (2021)	doaj.org/article/cedcdc9d16f541aaa07a5c67ef7f5d2b	Open access
32	Wang et al. (SHL) (2021)	doi.org/10.3389/fcomp.2021.713719	Open access
33	Wang and Jiang (2022)	doi.org/10.3390/s22176712	Open access
34	Tahir and Loo (2021)	doi.org/10.3390/healthcare9121676	Open access (also arXiv:2106.11776)
35	Dalakleidi et al. (2022)	www.ncbi.nlm.nih.gov/pmc/articles/PMC9776640/	Open access (PubMed Central)
36	Ismail and Sahari (2024)	penerbit.uthm.edu.my/periodicals/index.php/peat/article/view/14767	Open access
37	Miller et al. (2020)	arxiv.org/abs/2007.06933	Free on arXiv
38	Anand et al. (2021)	doi.org/10.1016/j.enbuild.2021.111478	Repository record: research-collection.ethz.ch (handle 20.500.11850/508529)
39	Himeur et al. (2021)	arxiv.org/abs/2010.04560	Free preprint on arXiv
40	Azad, Rajabi and Estebsari (2023)	arxiv.org/abs/2306.05017	Free on arXiv